AI-Optical Design Evolution
- Four Pillars -

Summary Table with Key References

Updated March 5, 2026

Pillar 1: 湧現 (Emergence)
Pillar 2: 湧創 (Continuous Innovation)
Pillar 3: 湧視 (Unaddressed Limits)
Pillar 4: 湧知 (Emergent Knowledge)

Original three pillars: **湧現** | **湧創** | **湧視**

Proposed fourth pillar: **湧知**

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1 Pillar 1: 湧現 (Emergence)

Pillar 1: 湧現 — Emergence

"Collective behavior of optical elements creates new functionalities not inherent in any single component."

#	Emergent System	Components	Emergent Behavior	Status (May 2025)	Status (Mar 2026)	Ref.
1	Computational Imaging	Coded aperture + Algorithm + Detector	Super-resolution beyond diffraction limit	Lab demos	**AI wavefront compensation: 40% → 98% accuracy recovery**	R19; R20
2	Metalens Arrays	Subwavelength structures	Anomalous dispersion control	Active research	**Achromatic RGB bilayer metalens: 80° FOV, NA=0.65**	R17
3	Adaptive Imaging	Deformable mirror + WFS + Control	Real-time aberration correction	Established	**DRL auto-tuning of photonic circuits in 21 iterations**	R9
4	Light Field Cameras	Microlens array + Sensor + Processing	Post-capture refocusing	Commercial	Mature; incremental	—
5	Quantum Imaging	Entangled photons + Coincidence detection	Ghost imaging, sub-shot noise	Early lab demos	**Metasurface SPDC: $>10^4\times$ resolution cells vs bulk crystal**	R2
6	**Quantum-Metasurface Multi-modal (NEW)**	Pol.-entangled source + Metasurface + ICCD	Phase contrast + edge + superimposed; self-calibration	Not identified	**Self-calibrated edge imaging via "off-state" noise subtraction**	R3
7	**AI-Optical Co-Design (NEW)**	LLM + Physics engine + RL + Zemax	Lens systems from natural language specs	Concept only	**OptiAgent: first LLM-agent for optical design with physics-driven RL**	R1
8	**Nonlinear Metasurface Quantum Sources (NEW)**	Metagrating on $LiNbO_3$ thin film	All-optical scanning of entangled photon emission angle	Not identified	**Demonstrated at IR wavelengths**	R2

2 Pillar 2: 湧創 (Continuous Innovation)

Pillar 2: 湧創 — Continuous Innovation

"Shifting from periodic breakthroughs to constant incremental improvements through real-time feedback loops."

#	Aspect	Discrete Evolution (2025)	Continuous Target (2025)	Current Status (Mar 2026)	Ref.
1	Design Cycle	6–24 months per design	Real-time adaptation (hours/days)	**Metasurface: hours (achieved); Classical: days (partial); Hybrid: weeks**	R5; R1
2	Feedback Integration	Post-production analysis	In-situ performance monitoring	**ML tolerance prediction replaces millions of ray-trace perturbations; "right first time"**	R11
3	Knowledge Transfer	Conference papers, patents	Shared AI models, cloud databases	**Open datasets: OptiDesignQA; Open tools: LensAI, LensRL; No unified platform yet**	R14; R15
4	Optimization	Fixed merit functions	Dynamic merit function evolution	**Still static; RL reward shaping (OptiAgent) closest but not self-evolving**	R1
5	Manufacturing Link	Sequential handoff	Concurrent design-manufacture	**Micro-lens + LED co-design in production (CES 2026); digital twin partial**	R12
6	**AI Agent (NEW)**	Not in original	—	**LLM agents with Zemax function calls in closed loop; RL on H100 GPUs**	R1; R18
7	**Fab-Aware Design (NEW)**	Not in original	—	**ML-designed lens: 90% prob. of meeting spec despite tolerances**	R11

3 Pillar 3: 湧視 (Unaddressed Limits)

Pillar 3: 湧視 — Unaddressed Physical Limits

"Physical phenomena that current lens design paradigms ignore, work around, or haven't yet exploited."

#	Phenom.	2025 Assess.	Untapped Potential	Status (Mar 2026)	Maturity Shift	Ref.
1	Quantum Noise	Seen as limitation	Information source	**Entangled photon pairs from metasurface SPDC; improved SNR**	Limitation → Resource	R2; R4
2	Thermodynamic Limits	Ignored	Energy optimization	**Still largely ignored; thermal meta-devices advancing separately**	No change	Chern GitHub Thermal Photonics trilogy
3	Vacuum Fluctuations	Not considered	Casimir effects	**Remains exotic; no integration**	No change	—
4	Nonlinear Dispersion	Avoided	Soliton benefits	**AI navigates complex parameter spaces; OptoGPT for multilayer**	Minimized → AI-navigated	R10
5	Spin-Orbit Coupling	Unknown	Geometric phase control	**Photonic skyrmions via GRIN + metasurface; vdW (8 μm hBN) vortex gen.**	Unknown → Active	R6; R7
6	Topology	Not exploited	Protected optical states	**Fab-compatible SOI spin-valley Hall phase; pseudo-spin framework**	Not Exploited → Design Tool	R8
7	Time Reversal	Static only	Phase conjugation	**In-situ RL for holographic phase; adaptive control advancing**	Static → Adaptive	R9

4 Pillar 4: 湧知 (Emergent Knowledge) — New Pillar

Pillar 4: 湧知 — Emergent Knowledge

"New optical design knowledge that emerges from AI-human collaboration, where the AI discovers principles no human designer would have conceived."

#	Knowledge Domain	How It Emerges	Example / Evidence	Implication	Ref.
1	AI-discovered design topologies	LLM generates valid configs not in training data	OptiAgent: novel prescriptions via RL; outperforms traditional + vanilla LLM	New starting points human designers wouldn't explore	R1
2	Inverse design insights	NNs solve inverse scattering; reveal structure-function links	OptoGPT: dual-band AR coating, fewer layers, fraction of a second	Simpler solutions exposing hidden design rules	R10
3	Tolerance-aware principles	ML predicts mfg. sensitivity; reveals dominant parameters	ML identifies lens spacings as dominant tolerance drivers	"Optimize then tolerance" → "tolerance-informed optimization"	R11
4	PINN discoveries	PINN enforces physics; residuals reveal unmodeled physics	Singlet optimization + Eikonal Bridge; residuals flag model breakdown	PINNs as "physics debuggers" for model completeness	R21; Chern GitHub trilogy;
5	Cross-domain transfer	AI trained on one domain transfers to another	RL on aspheric singlet transfers strategies to other lens types	"Optical design intuition" in AI weights, extractable as heuristics	R15
6	Quantum-classical boundary	Metasurface quantum imaging reveals classical model limits	Multimodal quantum imaging with self-calibration	Quantitative thresholds for when quantum treatment is needed	R3; R4
7	Structured light generation	Spin-orbit coupling in simple materials	8 μm hBN: OAM ± 2 vortex via spin-orbit, no nanostructure	"Material-intrinsic" wavefront shaping without fab complexity	R7

5 Cross-Pillar Linkage Map

The four pillars are not independent—they form a coupled feedback system.

Connection	From	To	Mechanism	Status
Engineered emergence requires accessing hidden physics	湧視	湧現	Metasurfaces access spin-orbit, topology, quantum DOF → new emergent behaviors	Active
Continuous innovation generates new knowledge	湧創	湧知	Fast iteration loops produce unexpected solutions → new design principles	Emerging
Emergent knowledge reveals new physical frontiers	湧知	湧視	AI-discovered designs expose physics we didn't know was relevant	Early stage
New physics enables faster innovation cycles	湧視	湧創	Topology provides robust, fab-tolerant designs → fewer iterations needed	Theoretical
Emergence creates knowledge that feeds back	湧現	湧知	System-level surprises become codified design principles	Demonstrated (quantum self-cal.)
Knowledge accelerates engineered emergence	湧知	湧現	Understanding why emergence works allows designing for it intentionally	OptiAgent is first example

6 Key Reference Index

All references cited across the four pillars, with pillar relevance indicated. Click any reference number to jump to its full entry in Section 7.

#	Reference	Year	Pillar Relevance
R1	Geng et al., "OptiAgent: A Physics-Driven Agentic Framework for Automated Optical Design," arXiv:2602.23761	2026	湧現 + 湧創 + 湧知
R2	Ma et al., "Quantum imaging using spatially entangled photon pairs from a nonlinear metasurface," eLight 5, 2	2025	湧現 + 湧視
R3	Zhu, Wang et al., "Metasurface-assisted multimodal quantum imaging," PNAS	2025	湧現 + 湧知
R4	"Quantum Imaging with Metasurfaces: Gains, Limitations, and Prospects," MDPI Photonics 13(1), 69	2026	湧視
R5	Penn State / Nanophotonics, "AI approach takes optical system design from months to milliseconds"	2026	湧創
R6	Lei et al., "Topological Quasiparticles of Light: Spin-Orbit Coupling to Photonic Skyrmions," Laser & Photonics Reviews	2025	湧視
R7	"Spin-orbit coupling in van der Waals materials for optical vortex generation," Light: Sci. & Appl.	2025	湧視 + 湧知
R8	Liu, Qiu & Yan, "Winding coupling phase for pseudo-spin-derived topological photonics," Nature Communications	2025	湧視
R9	"Model-free optical processors using in situ RL with PPO," Light: Sci. & Appl.	2026	湧創 + 湧知
R10	Ma, Wang & Guo, "OptoGPT: foundation model for inverse design in optical multilayer thin films," Opto-Electronic Advances 7, 240062	2024	湧知
R11	Yang et al., "Deep learning for polarization optical system automated design," Photonics 11	2024	湧創 + 湧知
R12	CES 2026 Optical and Lens Technology Depth Report, lensmanufacture.com	2026	湧創
R13	"AI Opens Up Optical System Design," Optica-OPN Vol. 37, January 2026	2026	All pillars
R14	LensAI, GitHub: HarrisonKramer/LensAI	Ongoing	湧創 + 湧知
R15	LensRL, GitHub: HarrisonKramer/LensRL	Ongoing	湧創
R16	ST Blog, "Top Tech Trends 2026: Metasurface technology shifts the paradigm"	2026	湧視
R17	Roadmap for Optical Metasurfaces, ACS Nano	2024	湧視 + 湧現
R18	Burkhardt et al., "Active Alignments of Lens Systems with Reinforcement Learning," arXiv:2503.02075	2025	湧創
R19	Hoshi et al., "Wavefront-aberration-tolerant diffractive deep neural networks using volume holographic optical elements," Scientific Reports 15, 1104	2025	湧現
R20	Yenra, "AI Optical System Design: 19 Advances (2025)," Review article	2025	湧現 + 湧創
R21	Raissi, Perdikaris & Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," Journal of Computational Physics, vol. 378, pp. 686–707	2019	湧知

7 Complete Reference List

Full bibliographic details for all references cited in the four-pillar tables. References are ordered by their in-text index (R1–R21). Click any reference number to return to the Key Reference Index.

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R2 J. Ma, J. Ren, J. Zhang, et al., "Quantum imaging using spatially entangled photon pairs from a nonlinear metasurface," *eLight*, vol. 5, art. 2, 2025. https://doi.org/10.1186/s43593-024-00080-8

R3 S. Zhu, Q. Wang, S. Wang, et al., "Metasurface-assisted multimodal quantum imaging," *Proceedings of the National Academy of Sciences (PNAS)*, May 2025. https://doi.org/10.1073/pnas.2500760122

R4 H. Z. Khan et al., "Quantum Imaging with Metasurfaces: Gains, Limitations, and Prospects," *Photonics (MDPI)*, vol. 13, no. 1, art. 69, January 2026. https://www.mdpi.com/2304-6732/13/1/69

R5 Penn State University, "AI approach takes optical system design from months to milliseconds," *Penn State News / Nanophotonics* (cover article, October 2025 issue), January 2026. https://www.psu.edu/news/engineering/story/ai-approach-takes-optical-system-design-months-millisec

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R7 (Authors), "Spin-orbit coupling in van der Waals materials for optical vortex generation," *Light: Science & Applications*, vol. 14, August 2025. https://doi.org/10.1038/s41377-025-01926-7

R8 T. Liu, M. Qiu, and W. Yan, "Winding coupling phase for pseudo-spin-derived topological photonics," *Nature Communications*, December 2025. https://doi.org/10.1038/s41467-025-67627-2

R9 (Authors), "Model-free optical processors using in situ reinforcement learning with proximal policy optimization," *Light: Science & Applications*, vol. 14, January 2026. https://doi.org/10.1038/s41377-025-02148-7

R10 T. Ma, H. Wang, and L. J. Guo, "OptoGPT: A foundation model for inverse design in optical multilayer thin film structures," *Opto-Electronic Advances*, vol. 7, art. 240062, 2024. https://doi.org/10.29026/oea.2024.240062

R11 S. Yang, M. Xu, H. Sun, et al., "Deep learning for polarization optical system automated design," *Photonics (MDPI)*, vol. 11, 2024.

R12 Shanghai Silk Optical Technology Co., Ltd., "CES 2026 Optical and Lens Technology Depth Report: Redefining Vision in the Age of Physical AI," January 2026. https://www.lensmanufacture.com/news-show-738.html

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R17 A. M. Shaltout et al., "Roadmap for Optical Metasurfaces," *ACS Nano*, 2024. https://nanoscience.ucf.edu/chanda/wp-content/uploads/sites/15/2024/03/2024_Roadmap-for-Optical-Metasurfaces.pdf

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R19 I. Hoshi, K. Wakunami, Y. Ichihashi, and R. Oi, "Wavefront-aberration-tolerant diffractive deep neural networks using volume holographic optical elements," *Scientific Reports*, vol. 15, art. 1104, 2025. https://doi.org/10.1038/s41598-024-82791-z

R20 Yenra, "AI Optical System Design: 19 Advances (2025)," Review article, 2025. https://yenra.com/ai20/optical-system-design/

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Note: For references marked "(Authors)" where only partial author information was available from search results, full author lists should be verified from the original publications.

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Original three pillars: **湧現** (Emergence), **湧創** (Continuous Innovation), **湧視** (Unaddressed Limits)

Proposed fourth pillar: **湧知** (Emergent Knowledge)